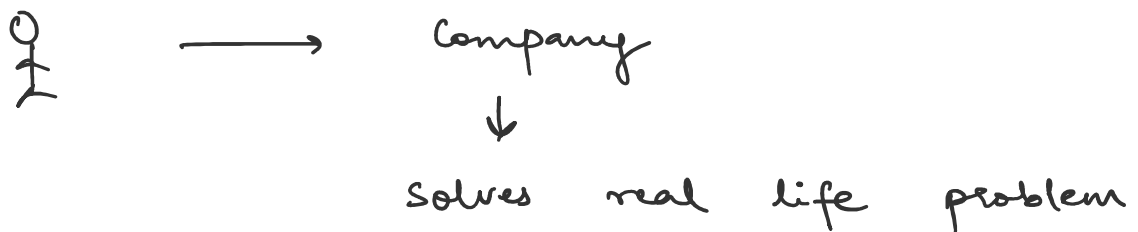


Class 1

Thursday, June 4, 2026 9:07 PM

Introduction to System Design



Social N/w Sites



Connected to friends → Instagram, Fb

E-Commerce websites



To shop online → Amazon

Groceries



Quick Delivery → Walmart

Banking



Banking App^s — Amex, Citi

Travel



Cab booking → Uber



Identify problem



Build scalable solⁿ

What s/w engg doing here —

→ Make things easier.

→ Saving time

→ Reducing manual effort.

→ Solve real life problem → Technology

fb → PHP

○ → one appⁿ

System Design

Planning how a s/w system will work today and how it will handle growth in future.

Main goals—

1. Easy to use at the beginning.

2. Able to handle more users later.

Steps involved in software development -

1. Understand the requirement.
2. Designing the system.
3. Write the code and testing.
4. Deployment of appⁿ
5. Improve and maintain the appⁿ over time.

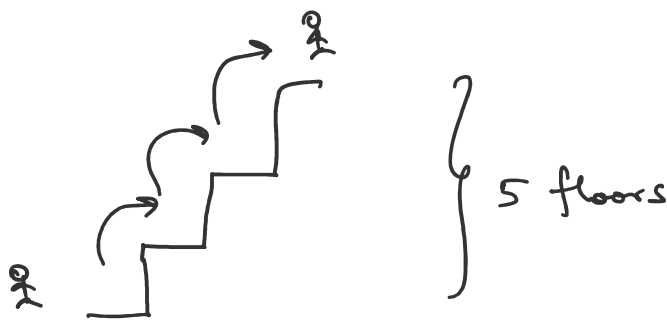
Requirement Gathering .



- Product owner | Product Mgr
- Customer (End users)
- Developer

Define the scope

- What features will be built first.
- What features will be built later.
- How the work will be divided over the time.



fb

version 1 -

→ user registration & login

version 2 -

→ Sending and accepting friend request.

version 3 -

→ Posting text updates.

version 4 -

→ uploading photos / videos

version 5 -

→ Liking and commenting.

Why scoping is important?

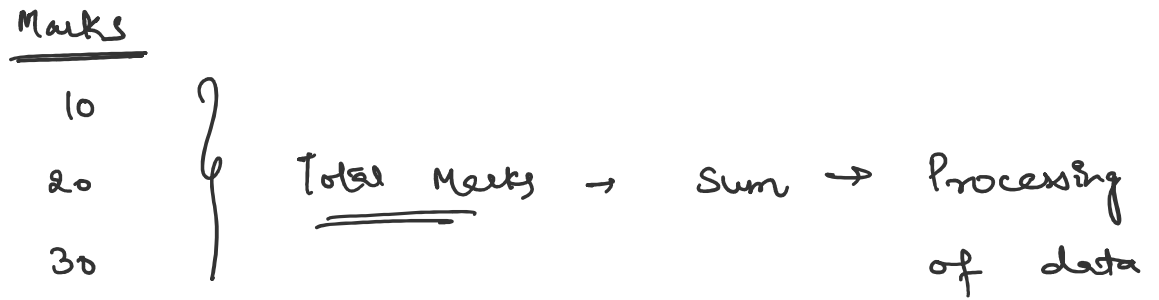
- set clear goals.
- Managing complexity
- Ensure steady progress.
- Better time mgt.

MVP — Minimum viable Product.

↓
for validation

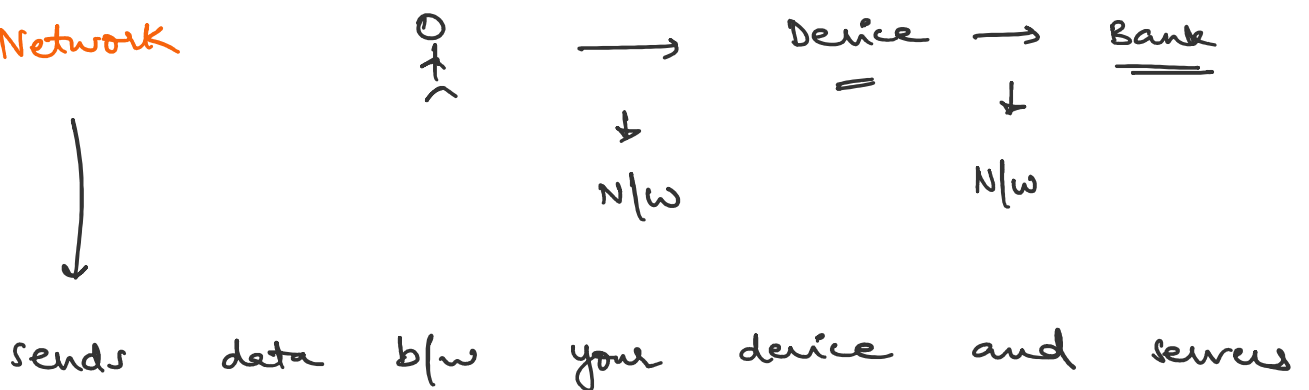
Small Demo | Prototype

Computation



[Bank statement → Feb 2026]

Network



SDLC → S/w Development Lifecycle

System Design

LLD



low level
Design

HLD



High level
Design

- ① Requirement Gathering
- ② Scoping
- ③ Estimation of data size
- ④ Components (Architecture)
- ⑤ Implementation of each components
- ⑥ Testing
- ⑦ Deployment

